

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

May 2025

Course Code: **PHY101R01**

Course: **ENGINEERING PHYSICS**

QP No. : **U141-2**

Duration: **3 hours**

Max. Marks: **100**

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. Obtain an expression for de Broglie wavelength associated with electrons.
2. The ultraviolet light of wavelength 300 nm is incident on a potassium surface. Calculate the maximum kinetic energy of photoelectron in eV. The work function of potassium is 2.2 eV.
3. Sketch the refractive index profile of step-index fiber and graded-index fiber.
4. Estimate the refractive index of core and cladding material of a fiber. (Given $NA=0.22$ and relative refractive index difference $\Delta=0.012$).
5. Recall four level lasers with diagram.
6. If magnetic field of 1800 A/m produces a magnetic flux of $3 \times 10^{-5} \text{ Wb}$ in an iron bar of cross-sectional area 0.2 cm^2 . Calculate its relative permeability.
7. Mention any two properties of ferroelectric materials.

8. Draw Frequency versus Polarisation plot for dielectric material.
9. Differentiate direct energy gap and indirect energy gap semiconductors.
10. For intrinsic silicon, the room temperature electrical conductivity is $4 \times 10^{-4} \Omega^{-1} m^{-1}$. The electron and hole mobilities are 0.14 and $0.040 m^2 V^{-1} s^{-1}$, respectively. Compute the electron and hole concentration at room temperature.

PART - B

Answer any Four questions

4 x 15 = 60 Marks

11. With neat instrument module discuss the working principle of scanning tunnelling microscope (STM). Also discuss different modes of operating STM.
12. a) What is mode in optical fiber? Based on the modes classify different types of optical fibers. (12)
- b) Find the diameter of the core for single-mode transmission at $8500 \text{ \AA}$ whose refractive indices for core and cladding are 1.48 and 1.47, respectively. (3)
13. Describe the construction and working of CO_2 molecular gas laser with energy level diagram. Mention its applications.
14. Based on the magnetic properties, classify magnetic materials.
15. a) Derive an expression for electronic polarisability and discuss its results. (14)
- b) Silicon has relative permittivity of 11.7. It contains $4.82 \times 10^{28} \text{ atoms/m}^3$. Calculate the dipole moment of each atom in a field of 10^4 V/m . (3)

- ✓ 16. a) What is Hall effect? Discuss the experimental determination of Hall coefficient with necessary diagrams. (12)
- b) The Hall coefficient of certain Si specimen was found to be $-7.35 \times 10^{-5} \text{ m}^3 \text{ C}^{-1}$ from 100 K to 400 K. Determine the nature of semiconductor if conductivity was found to be $200 \Omega^{-1} \text{ m}^{-1}$. Calculate the density and mobility of charge carrier. (3)

PART – C

Answer the following

1 x 20 = 20 Marks

17. a) Discuss 0D, 1D and 2D materials along with the density of states spectrum. (5)
- b) An Nd-YAG laser emits light at a wavelength of $1.063 \times 10^{-6} \text{ m}$. If the output power is 20 W, then how many photons are emitted in 10 min when the laser is in operation? (5)
- c) Differentiate soft and hard magnetic materials. (5)
- d) The electron and hole mobilities in InSb semiconductor are 6 and $0.2 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$, respectively. At room temperature, the resistivity of InSb is $2 \times 10^{-4} \Omega \text{ m}$. Assuming that the material is intrinsic, determine its intrinsic carrier density at room temperature. (5)
